Port B can be programmed in mode 0 & 1.
And in mode 1 lower bits of Port C of group B are used for handshaking signals.
• So, from the above discussion it is clear that how interfacing of peripheral devices is performed with the processor.

input/output device এর সাথে communication করতে হবে

The control word format of the 8255: imp

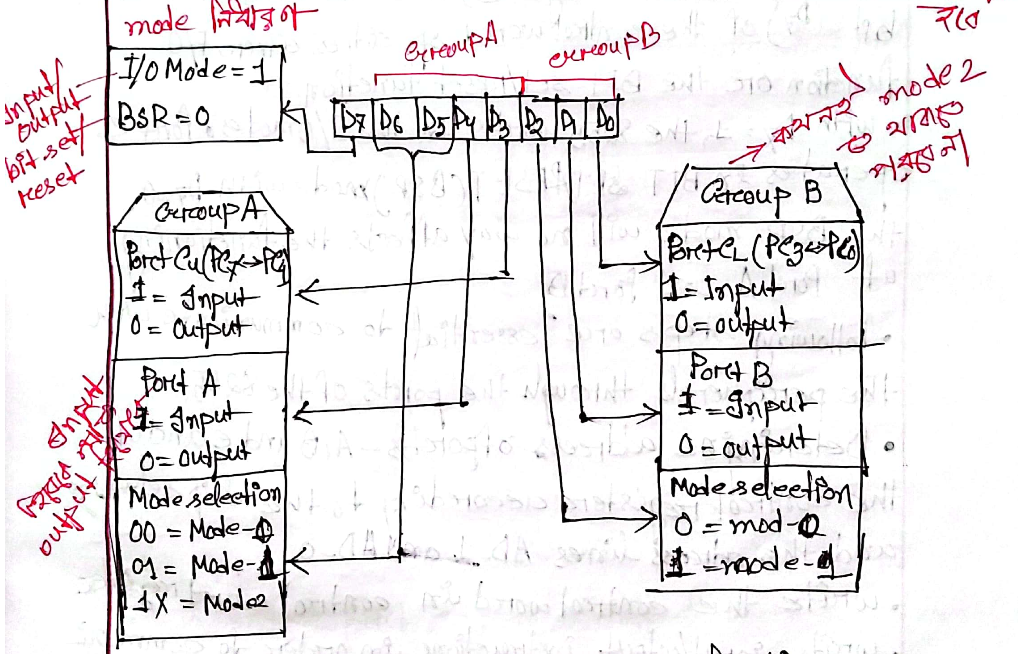


Fig: control word with group definition.

Imp

Control word Bits								Control Word	Port A	Port C (upper)	Port B	Port C (lower)
D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0					
1	0	0	0	0	0	0	0	80H	Output	output	output	output
1	0	0	0	0	0	0	1	81 H	Output	Output	Output	Input
1	0	0	0	0	0	1	0	82 H	Output	Output	Input	Output
1	0	0	0	0	0	1	1	83 H	output	output	Input	Input
1	0	0	0	1	0	0	0	88 H	Output	Input	output	output
1	0	0	0	1	0	0	1	89H	Output	Input	output	Input
1	0	0	0	1	0	1	0	8A H	output	Input	Input	output
1	0	0	0	1	0	1	1	8B H	Output	Input	Input	Input
1	0	0	1	0	0	0	0	90 H	Input	Output	Output	Output
1	0	0	1	0	0	0	1	91 H	Input	Output	Output	Input
1	0	0	1	0	0	1	0	92 H	Input	output	Input	output
1	0	0	1	0	0	1	1	93 H	Input	output	Input	Input
1	0	0	1	1	0	0	0	98 H	Input	Input	output	output
1	0	0	1	1	0	0	1	99 H	Input	Input	output	Input
1	0	0	1	1	0	1	0	9A H	Input	Input	Input	output
1	0	0	1	1	0	1	1	9B H	Input	Input	Input	Input

I/O mode

port A mode 0

port B mode 0

Imp

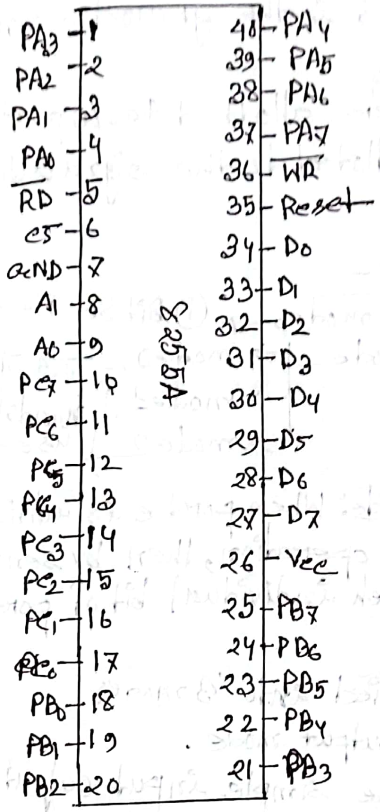
• Obtain the control word when the ports of 8255 A are to be used in mode 0 with port A as output and port B as input port, C as output port.

Soln:

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
1	0	0	0	0	0	1	0
			A	C(upper)		B	C(lower)

= 82H

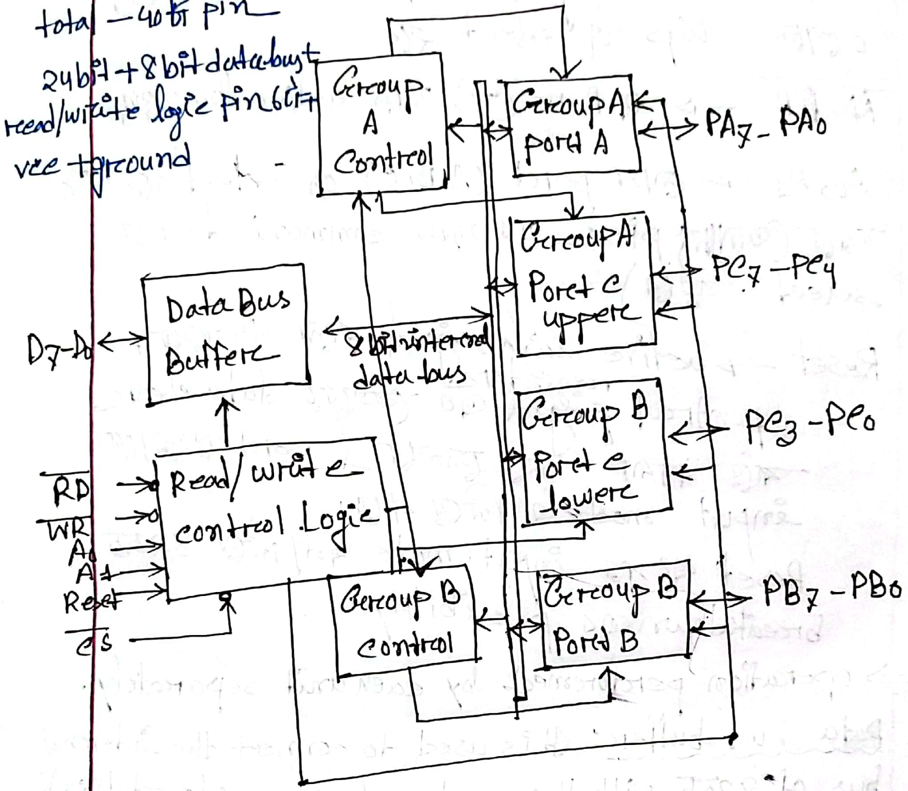
40 pin configuration of 8822 PPI



• As we see pin number 27 to 34 is allocated to data bus. These are tri-state data bus lines of bidirectional nature that connects 8255 with the processor. These are used to allows flow of data and control of status word between the 8255 and the processor.

Architectural representation of 8255 PPI:

total — 40 টা pin
24 bit + 8 bit databus
read/write logic pin 6 টা
vcc + ground



Processor এর System bus ~~আর~~ 8255 interenal data bus এর সাথে communication stablish করে, Data bus buffer + syncronization + interfacing করবে।

→ data transfer ~~হয়~~
→ data ~~that~~ transfer initialization + control + finish

read/write control logic:
~~কখন~~ read/write control logic কখন, কতটুকু সময়, নির্দিষ্ট moment এ read/write হবে, কতটুকু সময় পর থাকে, clock বন্ধ হবে সেটা control করে